

Postgraduate Studies in Business Analytics and AI

Introduction to the Programme

Jochen De Weerd

<https://bit.ly/session1-jochen>

2026-2027



Scan to register
your attendance!

Welcome to the 15th edition

- Our story
 - The programme started in the 2016–2017 academic year. Seppe vanden Broucke and I developed it together, almost like our own small start-up.
- Our academic home
 - LIRIS research group (Research Centre for Information Systems Engineering)
 - Department of Decision Sciences and Information Management
 - Faculty of Economics and Business, KU Leuven
 - 10 professors, 3 postdocs, ~20 PhD researchers

A fresh start

- Relaunched after a one-year pause
- New name: *Postgraduate Studies in Business Analytics & AI*
- A more balanced learning format
 - Ten session days spread across six months
 - Preparation before each session day
 - One coordinating lecturer per session day
 - Shorter teaching blocks on Friday
- A streamlined evaluation
 - One integrated case study dissertation
 - Final submission in June



1975 – 2025

In memory of Prof. Bart Baesens

The core team



Prof. Jochen De Weerd



Prof. Wouter Verbeke



Prof. Johannes De Smedt



Prof. Seppe vanden Broucke

About me

- Professor at KU Leuven
- PhD in Business Economics (2012)
- Postdoc at the IS School, Queensland University of Technology (QUT), Brisbane (2012-2013)
- Academic Coordinator of the Postgraduate Studies in Business Analytics and AI
- Research expertise: process and business analytics

My research team

Former researchers:

 Pieter De Koninck (2019) Process mining – trace clustering <i>Team Lead AI at Silverfin</i>	 Sandra Mitrovic (2019) Network analytics for churn prediction <i>Postdoc at IDSIA</i>	 Daria Bogdanova (2020) Co-promotor Automated feedback in SLEs <i>Educational Technologist at KU Leuven</i>	 Galina Deeva (2021) Learning analytics – Process mining <i>Behavioral Scientist at KBC</i>
 Rafael Van Belle (2023) Network analytics for fraud detection <i>ML Engineer at Dataroots</i>	 Björn Rafn Gunnarsson (2023) Predictive Process Monitoring with LSTMs <i>Data Scientist at NATO</i>	 Hans Weytjens (2023) Uncertainty for Predictive Process Monitoring <i>Senior Researcher at TUM</i>	 Jari Peepkorn (2023) Predictive Process Monitoring generalization & conformance <i>Postdoc at LIRIS</i>
 Carlos Eduardo Ortega Vázquez (2023) Imbalanced learning for fraud detection <i>Data Scientist at Telenet</i>	 Yannis Bertrand (2024) Co-promotor IoT process mining <i>Assistant Professor at UHasselt</i>	 Philipp Borchert (2024) NLP for business analytics <i>AI Research Engineer at Huawei</i>	 Margot Geerts (2025) Geospatial analytics for real estate valuation <i>Data Scientist at KBC</i>

Currently supervising:

 Brecht Wuyts Predictive Process Monitoring	 Xiaomeng He Intercase Featurization for Predictive Process Monitoring	 Aurélie Leribaux Predictive Process Modeling	 Victoria Sonnemans Prescriptive Process Monitoring
 Romain Bérard Prescriptive Process Mining	 Yongbo Yu Process Model Forecasting	 Zahra Ahmadi IoT process mining	 Jakob De Moor Prescriptive Process Monitoring

Session overview

1. Participant introduction
2. Practical details
3. Introduction to business analytics & AI
4. The data analytics process
5. Challenges
6. Tooling

1. Participant introduction

Getting to know each other

- Your one minute of fame
- Please tell us:
 - Who are you?
 - What do you do professionally?
 - What motivated you to join the programme?

2. Practical details

Programme overview

Day	Date	Topic	Lecturer
Day 1	25 Sep 2026	Introduction and data preprocessing	Jochen De Weerd
Day 2	9 Oct 2026	Supervised and unsupervised learning	Johannes De Smedt
Day 3	23 Oct 2026	Causal machine learning	Wouter Verbeke
Day 4	13 Nov 2026	Deep learning	Seppe vanden Broucke
Day 5	27 Nov 2026	Text and web mining	Matthias Bogaert
Day 6	11 Dec 2026	Large language models	Johannes De Smedt
Day 7	15 Jan 2027	Process mining	Jochen De Weerd
Day 8	29 Jan 2027	Reinforcement learning	Wouter Verbeke
Day 9	19 Feb 2027	Generative and agentic AI	Seppe vanden Broucke
Day 10	5 Mar 2027	Organisation, management and governance	Jochen De Weerd

Daily schedule

- Session 1 (13:00–15:00): Introduction and theory

Coffee break

- Session 2 (15:30–17:00): Hands-on exercises using your own laptop, Google Colab or a similar platform

Coffee break

- Session 3 (17:30–19:00): In-depth exploration, guest lecture or additional content

- Networking dinner (19:00–20:30)

Course documents

- We move away from printouts to only digital content
- Toledo
 - LMS of KU Leuven: <https://toledo.kuleuven.be/english>
 - This program has a dedicated community in which I will post all course documents electronically (also all additional information and links)
 - Toledo is also the place to be if you want to watch a recording of a session
 - You also find all practical information about the dissertation in this community
 - I also provided two books that can serve as refresher for some basic statistics (not mandatory)

Wifi access

- Network name: “campusroam” or “eduroam”
- Username: (Student ID)@kuleuven.be (Student ID = r-, m- or s-number)
- Password: the password that belongs to your user id
 - Note that you will need to first create your account with the information that is provided to you in the envelope with your student ID card
- Security: WPA2 Enterprise / Encryption: AES

Participant feedback

- Online questionnaire
 - Every session day, we will work with an online questionnaire to obtain your feedback on the program
 - Please participate!
 - Of course, immediate oral feedback is also more than welcome
- Make use of the breaks to approach myself or any of the other lecturers with your questions

Evaluation

- How to obtain the postgraduate certificate?
- Individual dissertation
 - I.e., a written document of anywhere between 5.000 and 15.000 words (10 – 30 pages)
 - A template is available (link in the guidelines + on Toledo)
 - Builds on the content of the program
 - Must involve data (real or artificially generated)
- Guidance by one of the program's lecturers
- Oral presentation (15 minutes presentation + 10 minutes Q&A)
- Presence in at least 8 of the 10 session days is required (attendance registration with QR code)
- Due to confidentiality, I cannot share example works of past participants

Timeline

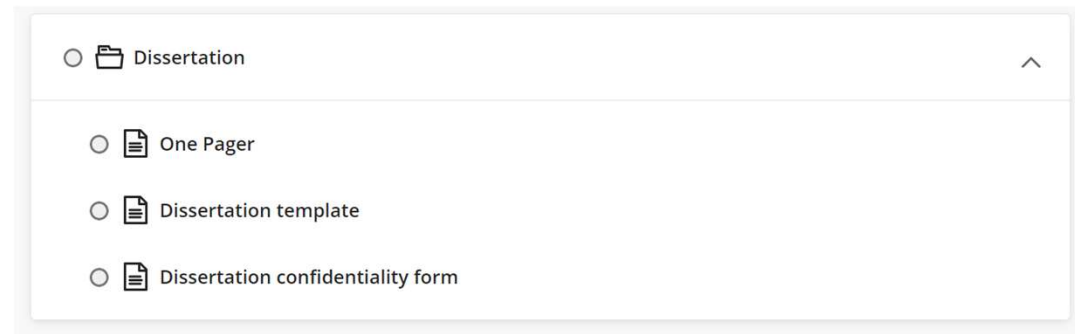
- One-pager deadline: **February 17, 2027 (23:59)**
 - One-pager template available
- Dissertation deadline: **June 9, 2027 (23:59)**
 - Submit pdf via email
- Presentation: **14–25 June, 2027**
- 2nd opportunity: **Wednesday September 1, 2027 (23:59)**

What if you fail to make the September 1 deadline?

- Further extension is not possible
 - End of the academic year
- If you still want to complete the dissertation, you will need to re-enroll for a new academic year
 - New minimal enrolment fee of €550

Advice and confidentiality

- Advice
 - Of course, your adviser will not do the work for you
 - Given our own busy agendas, we strongly prefer communication over email
- Confidentiality
 - We don't need to get access to your data!
 - Your submitted document is kept strictly confidential (only the two evaluators will have access)
 - There is a confidentiality form that you can sign covering the document, presentation, and any other information you share with us.

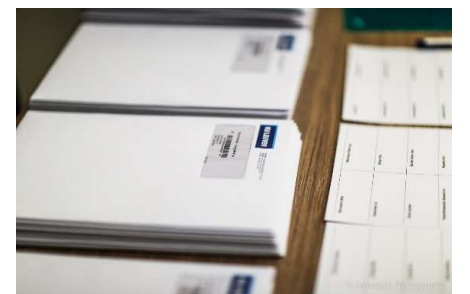


Presentation

- Online (TBD)
- Timing
 - Max. 15 minutes of presentation (+/- 20 slides)
 - 10 minutes of questions
- Further remarks
 - You can use a slide template of your choice
 - You don't need to send us your slide deck in advance
 - Don't include (parts of) slides that were part of our material

Success!

- We score your dissertation out of 20.
- I will inform you about your results soon after the last presentation slot
- Official results (via KU Leuven – ISP) will only be available beginning of July
- If successful, graduation ceremony in September!



Questions?

3. Introduction to business analytics & AI

Setting the scene

- Data contains value and knowledge
- People still claim “data is the new oil” (but I don’t agree)
- But to extract this knowledge, you need to be able to
 - Store it
 - Manage it
 - Analyze it
- *Business Analytics ≈ Data Mining ≈ Big Data ≈ Data Analytics ≈ Data Science ≈ Machine Learning ≈ AI ?*



“

What is AI?

What is it all about?

- AI = field of computer science dedicated to solving problems which *otherwise require human intelligence*
 - For example: image recognition, language understanding, learning from data, etc.
- Machine learning (ML) = the scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead
 - *Subset* of artificial intelligence
 - Machine learning algorithms build a mathematical model based on sample data, known as 'training data', to make predictions or decisions without being explicitly programmed to perform the task

What is it all about?

Artificial intelligence

Systems that perform tasks that normally require human intelligence

Whole programme · Day 10: organisation, management and governance

Machine learning

Learning patterns from data instead of programming rules

Day 2: supervised & unsupervised · Day 3: causal ML · Day 8: reinforcement learning

Most business analytics

Classic ML on curated, tabular data:
predicting churn, fraud, prices, demand, ...

Data preparation: Day 1

Process mining: Day 7

Deep learning

Neural networks with many layers

Day 4 · Day 5: text and web mining

Generative AI

Foundation models, LLMs, agents

Day 6 · Day 9

And what about statistics?

Statistics



- Describing data (average, standard deviation, ...)
- Quantitative modeling of distributions
- Quantitative modeling of relationships between variables
- Models often parametric and meant for human understanding

ML



- Algorithms for pattern extraction and modeling
- Many of which utilize statistical underpinnings
- Models often non-parametric and black box, meant to optimize predictive power

AI



- At first (1960s and later): programming paradigms to make machines appear intelligent
- 1980: neural network and SVM based approaches, true intelligence in reach, then lost
- 2010: deep learning; these are ML algorithms as well but often applied in difficult, non-traditional settings (computer vision, text) where simpler statistical approaches failed
- 2017: transformers
- 2022: foundation models and ChatGPT
- 2025: agents

We focus on ML, the heart of data analytics

- Extracting useful patterns and mathematical models from data
 - Instead of hand-coding, let the data speak
 - To help predict something, explain something, decide something
- Using:
 - 1) Data
 - 2) An algorithm
 - 3) A purpose
- Which are:
 - **Valid**: hold on new data with some certainty, i.e. generalizable
 - **Useful**: should be possible to act on the item, i.e. actionable
 - **Unexpected**: non-obvious to the system, i.e. interesting
 - **Understandable**: humans should be able to interpret the pattern, i.e. explainable

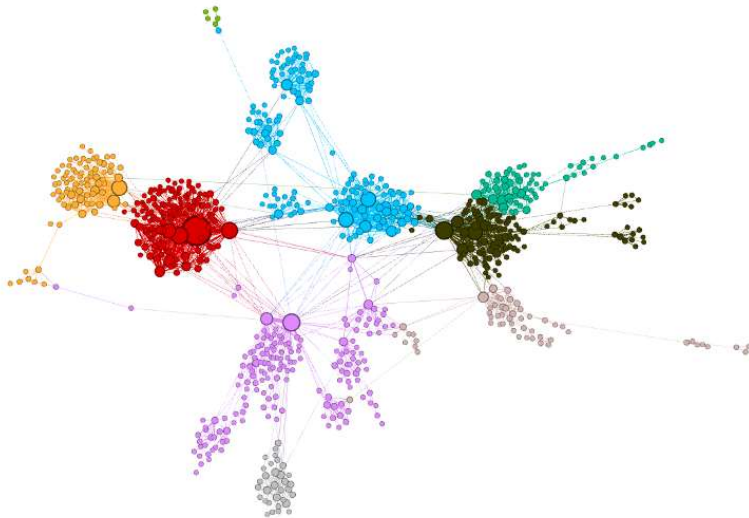
Why don't we ask Claude?

- You could, but ...
- For many tasks, zero-shot or few-shot prompting does not yet reach the performance of a dedicated model trained on a curated dataset
- Example: house price prediction (Geerts et al., 2025)
 - GPT-4o-mini and Llama 3 with zero- and few-shot prompting vs. gradient boosting (LightGBM) in four housing markets
 - The LLMs come close but remain less accurate: e.g. King County, MAPE 18.6% vs. 13.8%
 - Their prediction intervals are overconfident: 35–64% coverage vs. 85–90% for LightGBM (target 90%)
 - They overlook the location features that the dedicated model finds most important
- A dedicated model is also cheaper, faster, reproducible and easier to validate
- So we still need machine learning, and a carefully built data pipeline

Geerts, M., Reusens, M., Baesens, B., vanden Broucke, S., & De Weerd, J. (2025). On the performance of LLMs for real estate appraisal. arXiv:2506.11812

1) Data

Name	Age	Income	Occupation	...	Fraud y/n
Anne	34	1400	Midwife	...	N
Bert	53	1500	NA	...	N
Cyrus	23	1200	Office clerk	...	N
Dean	35	1500	Professor	...	Y



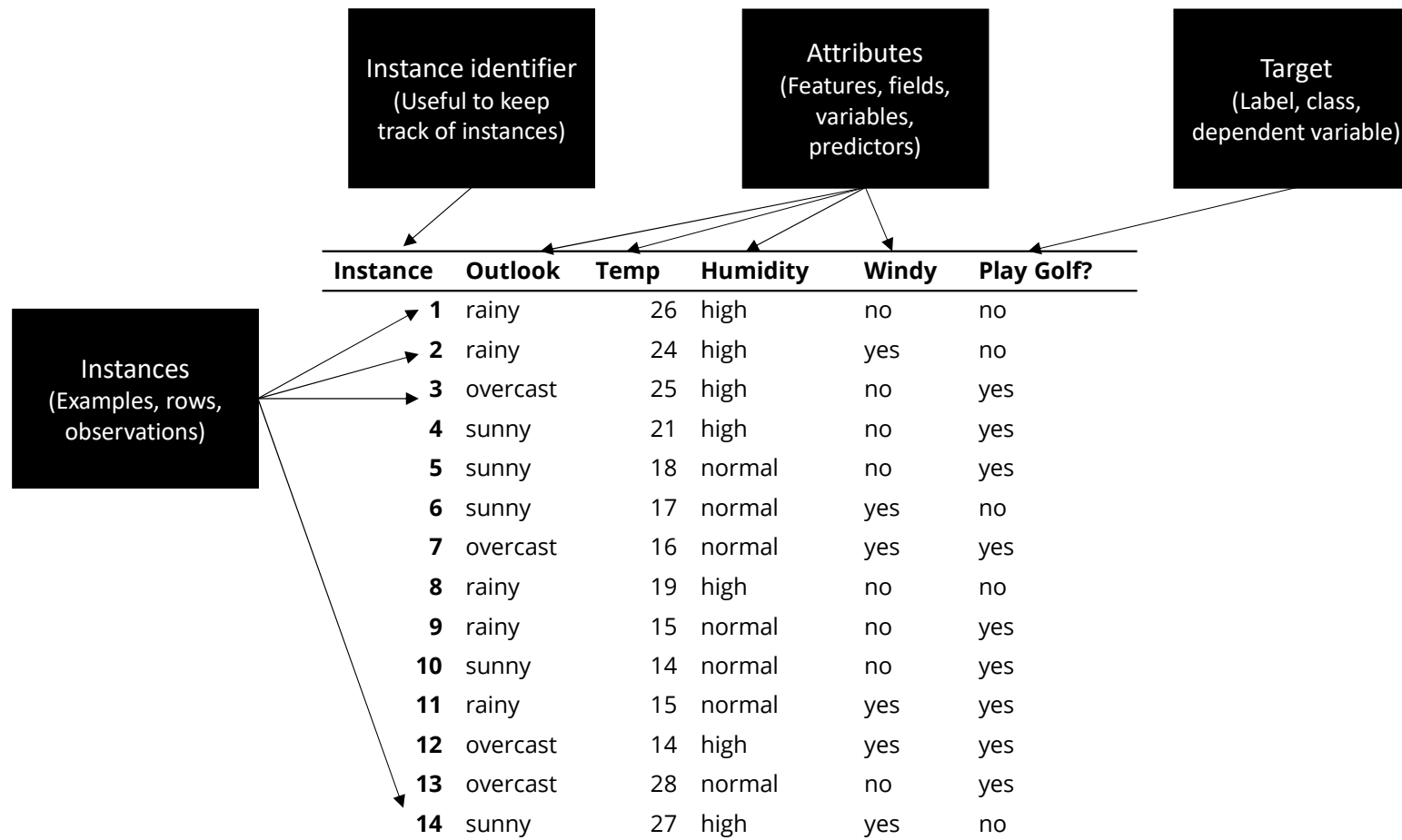
by ent270 ,ent223 updated 9:35 am et ,mon march 2 ,2015
 (ent223) ent63 went familial for fall at its fashion show in
 ent231 on sunday ,dedicating its collection to ``mamma``
 with nary a pair of ``mom jeans`` in sight .ent164 and ent21 ,
 who are behind the ent196 brand ,sent models down the
 runway in decidedly feminine dresses and skirts adorned
 with roses ,lace and even embroidered doodles by the
 designers 'own nieces and nephews .many of the looks
 featured saccharine needlework phrases like ``I love you ,
 . . .



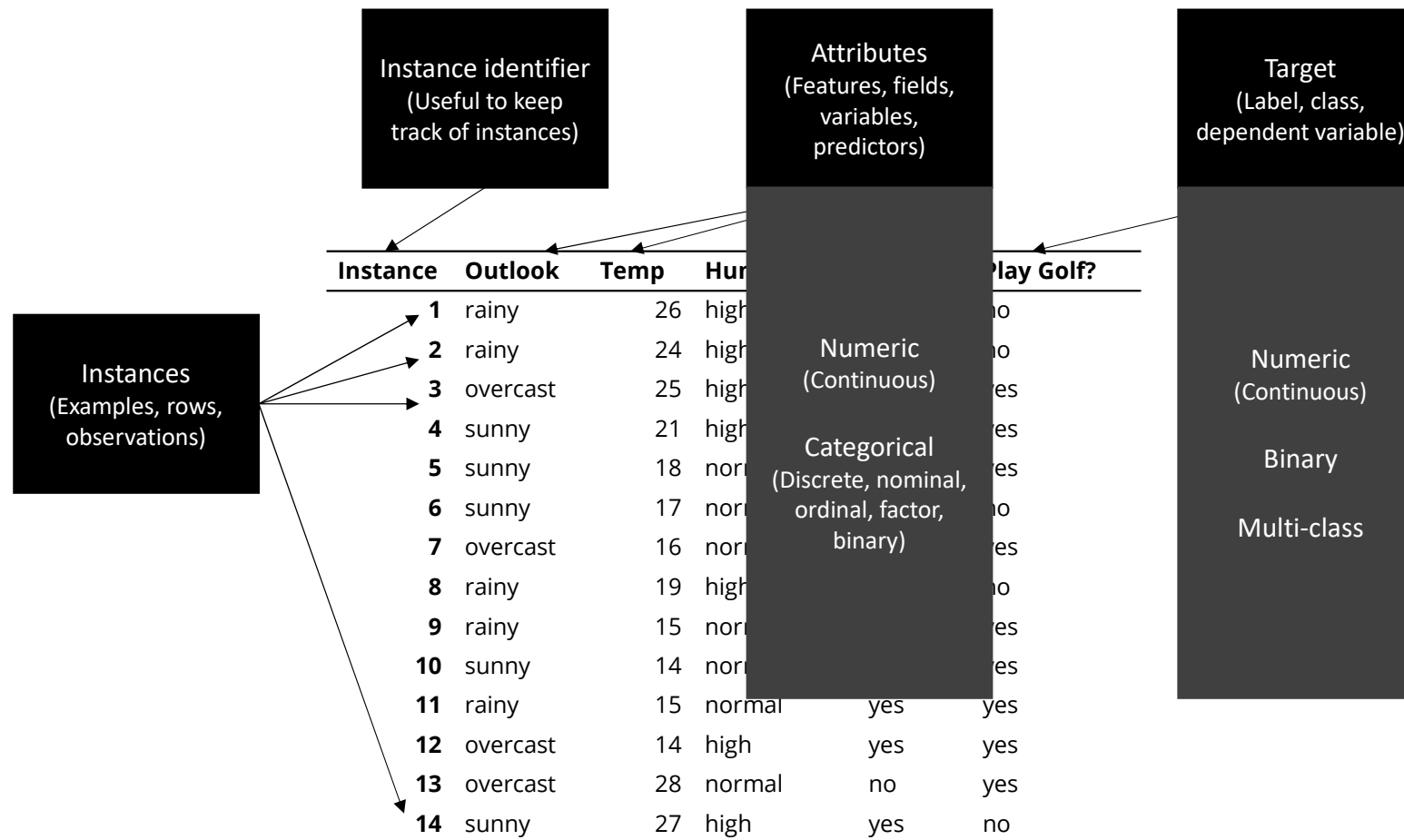
Structured vs. unstructured

- Structured vs. unstructured?
 - Tabular vs. relational, text, imagery, audio, ...
- Two main approaches to deal with non-tabular data
 - Making it tabular (“featurization”)
 - Using models that can directly utilize data as-is (“deep learning”)

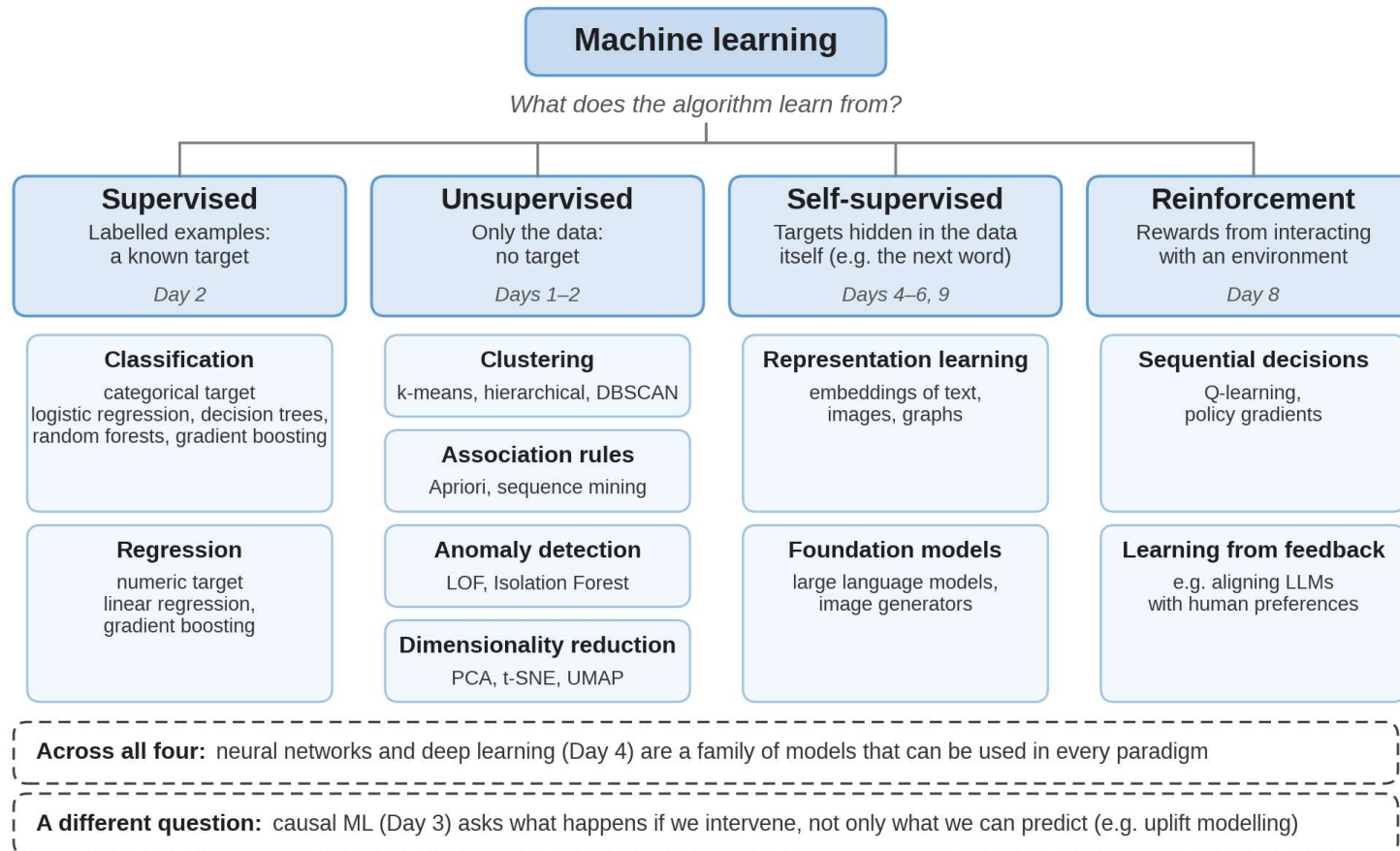
Datasets, attributes and instances



Datasets, attributes and instances



2) Algorithms

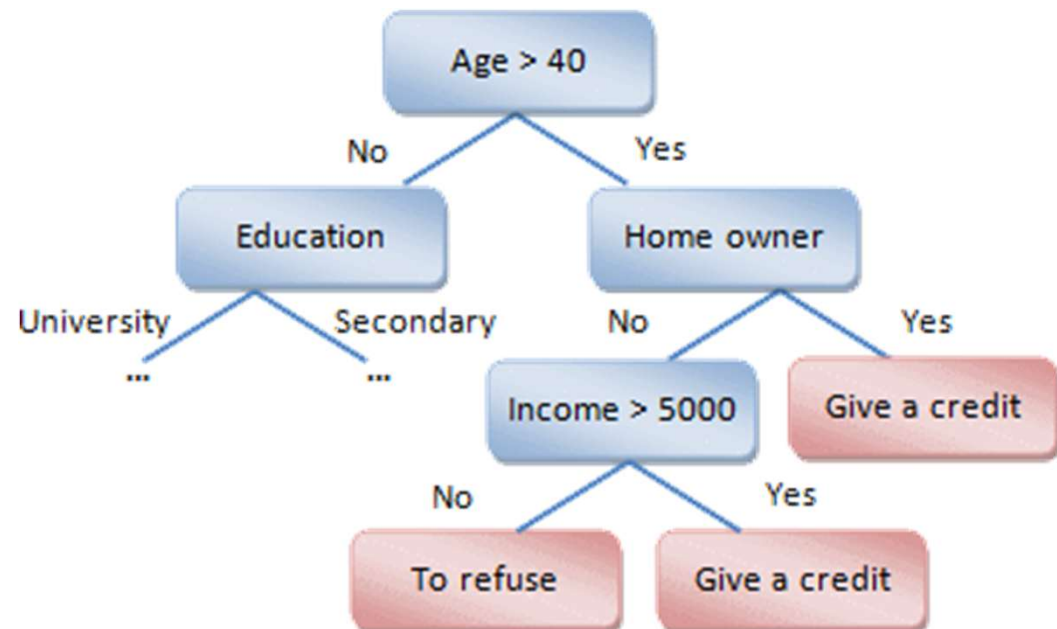


Supervised learning

- **You need a target/label!**
 - Classification: categorical target
 - Binary classification
 - Multiclass classification
 - Ordinal classification
 - Multilabel classification
 - Regression: continuous (numerical) target
 - Absolute values
 - Delta values
 - Quantiles regression

Example: decision tree

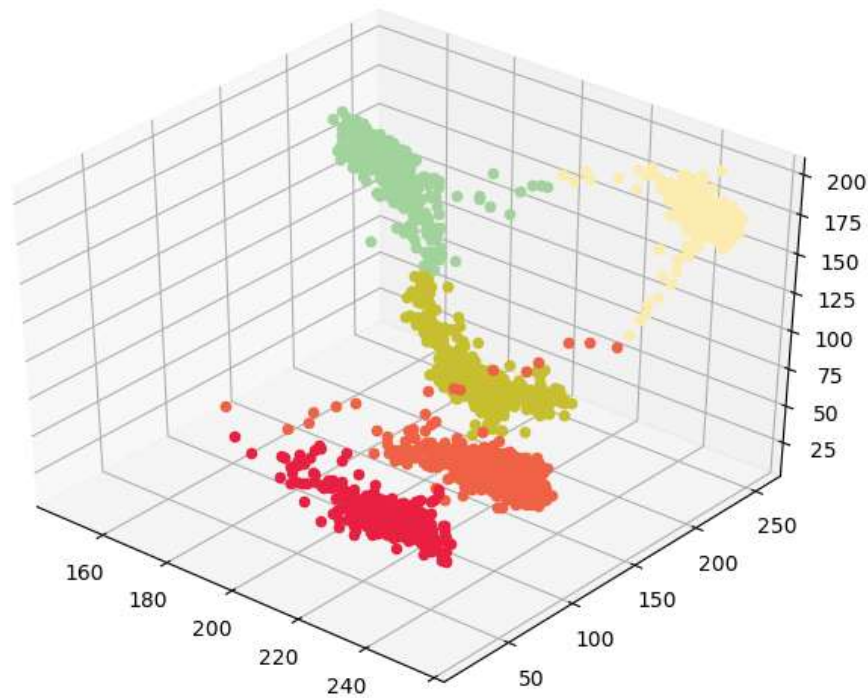
Using the data!



Unsupervised learning

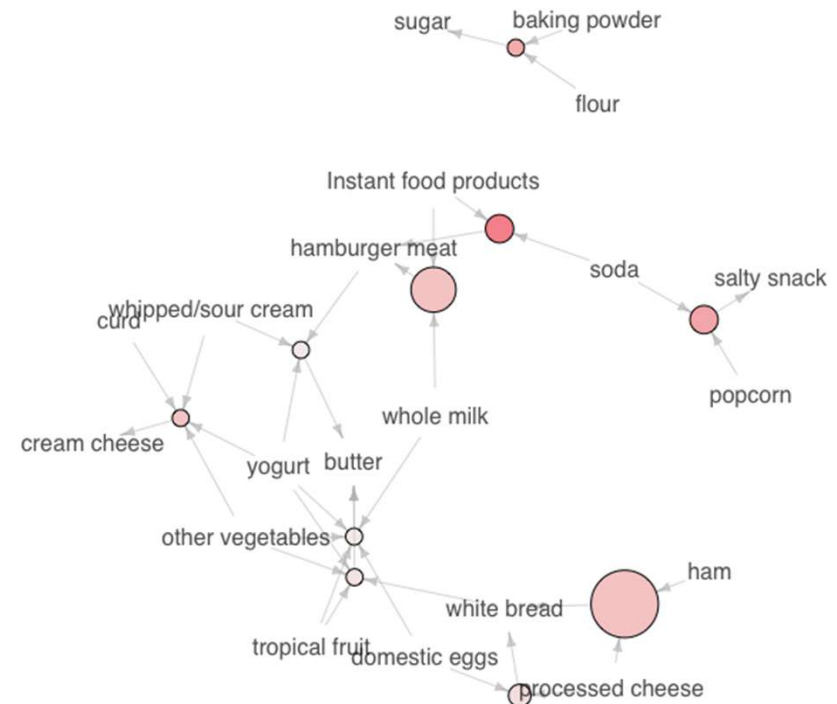
- **Extract patterns from the data as is**
 - Clustering: construct groups over the data set
 - Association/sequence/... rule mining: find rules of antecedents and consequents that describe the data
 - Anomaly detection: find outliers in the data set
 - (Dimensionality reduction: from many variables to fewer)

Clustering & Association rules



Graph for 10 rules

size: support (0.001 - 0.002)
color: lift (11.279 - 18.996)



3) Purpose

- Unsupervised learning techniques for descriptive analytics and supervised learning techniques for predictive analytics?
 - Yes, but...

WHAT IS YOUR BUSINESS QUESTION?

WHAT IS THE BUSINESS PROBLEM?

Purpose

- **Exploratory:** plots, distributions, quick charts, basic correlations – very visual
 - But who says you couldn't build a supervised model to help here?
- **Descriptive:** unsupervised – clustering, association rules
 - Depends on which style of descriptions you want to get; very often you already have some hypothesis going on
- **Explanatory:** unsupervised again?
 - Depending on target definition and model type used, a supervised model can be used as an explanatory means with not much generalization power going forward
- **Predictive:** supervised for sure (right?)
 - Though in many cases unsupervised techniques can be used here as a featurization or pre-processing step
- **Prescriptive:** “what should I do”
 - What-if analysis using a supervised model, or using good ole' operations research

Purpose

- Purpose is key!
 - Always a good question to ask: assume I have a model everyone's happy with; how will the model be used? Which features do I need to give it? Do I want it to make predictions going forward? What is my end goal?
- While ML is a powerful tool, it's not the solution for every problem!



"ML can help automate your processes, but not all automation problems require ML"

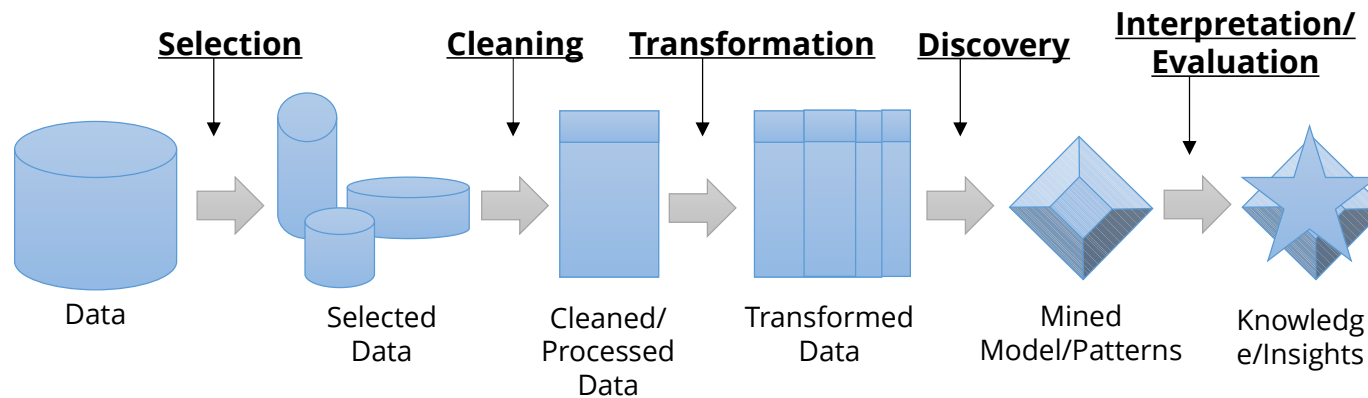
Purpose

- Why does AI/ML matter for business?
 1. **Automation:** It can help automate repetitive tasks, freeing up human workers for more strategic roles
 2. **Efficiency:** AI-driven insights help improve operational efficiency (e.g., predictive maintenance in manufacturing)
 3. **Customer experience:** Personalization and chatbots that enhance customer service
 4. **Data-driven decisions:** ML helps analyze vast datasets and derive actionable insights (e.g., fraud detection)

4. The data analytics process

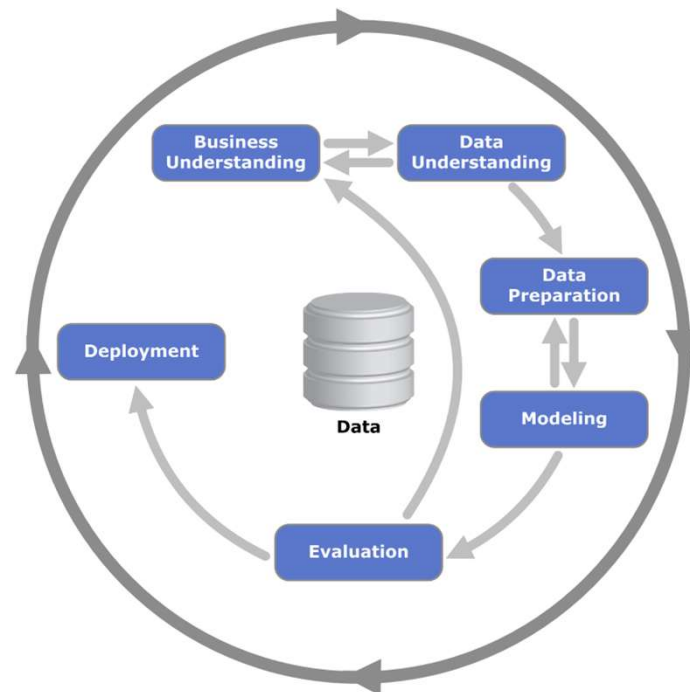
The data analytics process

- KDD process (“knowledge discovery in databases”)

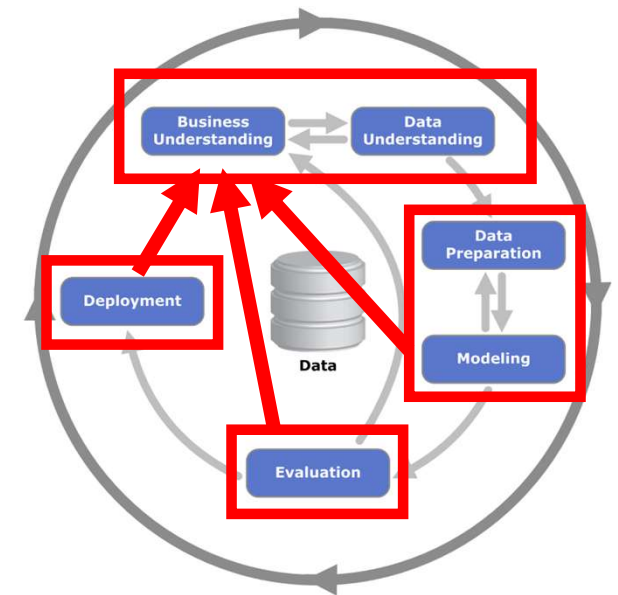
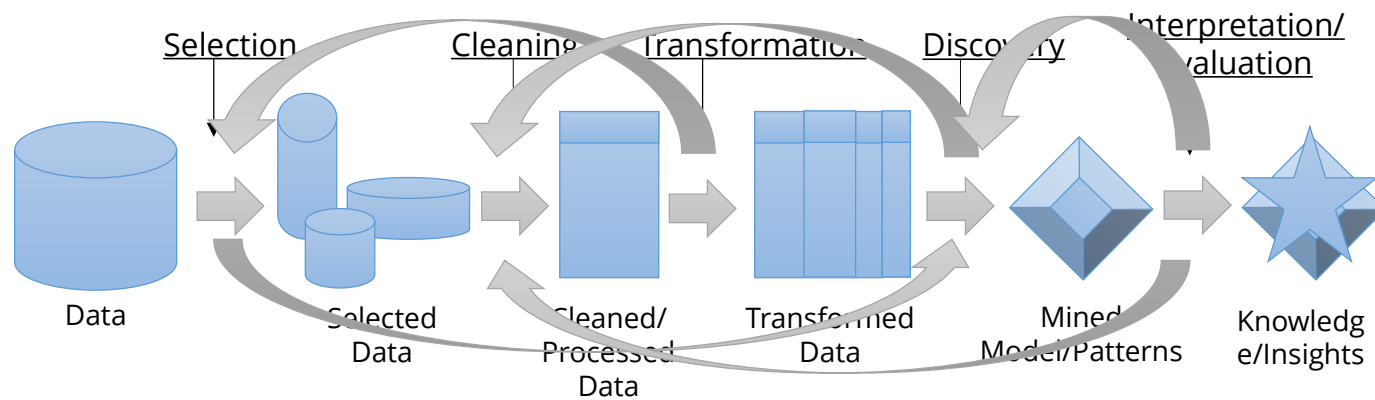


The data analytics process

- CRISP-DM (“cross-industry standard process for data mining”)



The real data analytics process



Managing AI projects

- AI projects are experiments, not construction projects
 - Plan for learning: iterate, and reassess viability at regular points
 - “Done” means a model that creates value in use, not a model that reaches an accuracy target
- Use a process model made for machine learning
 - CRISP-ML(Q): CRISP-DM with quality assurance in every phase, plus monitoring and maintenance after deployment
 - PMI-CPMAI: the Project Management Institute’s methodology for AI projects (iterative, data-first)
- Plan deployment, monitoring and governance from the start
 - E.g. ISO/IEC 42001 (AI management systems), EU AI Act (see Day 10)
- Foster AI power couples
 - Bring together a business expert with an AI expert

Vial, G., Cameron, A.-F., Giannelia, T., & Jiang, J. (2023). Managing artificial intelligence projects: Key insights from an AI consulting firm. *Information Systems Journal*, 33(3), 669–691.
Studer, S., et al. (2021). Towards CRISP-ML(Q): A machine learning process model with quality assurance methodology. *Machine Learning and Knowledge Extraction*, 3(2), 392–413.

Why AI projects fail

- More than 80% of AI projects fail, twice the rate of IT projects without AI
 - Based on interviews with 65 data scientists and engineers
- Five root causes
 1. The wrong problem or metric: business and technical teams are not aligned
 2. Not enough good data, or not enough domain knowledge about it
 3. Technology first: teams chase the latest technique instead of the user's problem
 4. Too little investment in data infrastructure and deployment
 5. The problem is beyond what current AI can do

Ryseff, J., De Bruhl, B. F., & Newberry, S. J. (2024). The root causes of failure for artificial intelligence projects and how they can succeed. RAND Corporation, RR-A2680-1.

The same pattern with generative AI

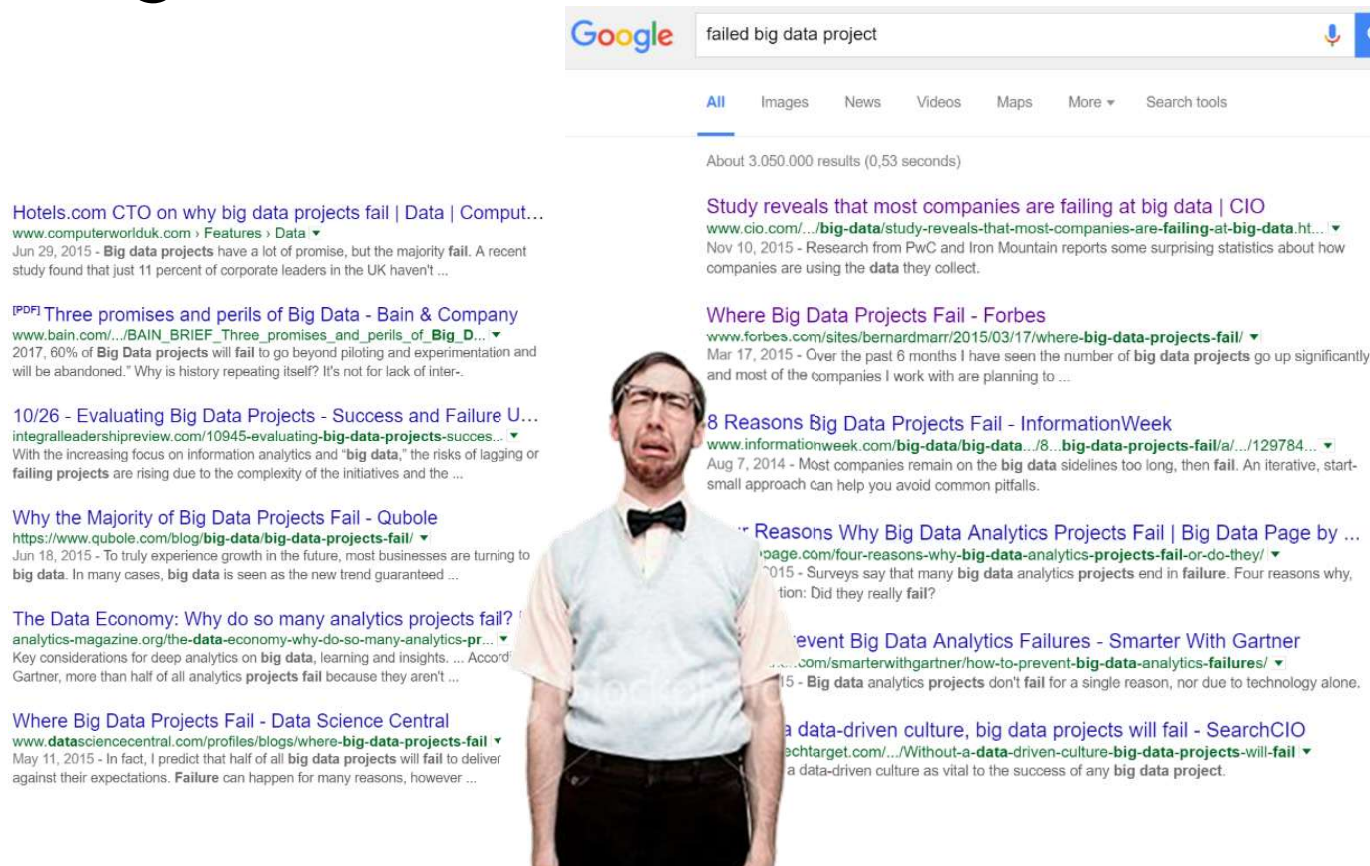
- 95% of enterprise GenAI pilots show no measurable impact on profit and loss (MIT NANDA, 2025)
 - The method is debated: read it as a signal, not as a precise figure
 - Main gap: tools that do not learn from feedback or fit into existing workflows
- Over 40% of agentic AI projects will be cancelled by the end of 2027 (Gartner, 2025)
 - Reasons: escalating costs, unclear business value, inadequate risk controls
 - “Agent washing”: existing chatbots and automation tools rebranded as agents
- The root causes are the same as for classic machine learning
 - Clear business problem, good data, investment in deployment, realistic expectations

MIT NANDA (2025). The GenAI divide: State of AI in business 2025.

Gartner (2025, June 25). Gartner predicts over 40% of agentic AI projects will be canceled by end of 2027 [Press release].

5. Challenges

Challenges



Google

failed big data project

All Images News Videos Maps More Search tools

About 3,050,000 results (0,53 seconds)

Hotels.com CTO on why big data projects fail | Data | Comput...
www.computerworlduk.com/Features/Data
Jun 29, 2015 - Big data projects have a lot of promise, but the majority fail. A recent study found that just 11 percent of corporate leaders in the UK haven't ...

[PDF] Three promises and perils of Big Data - Bain & Company
www.bain.com/.../BAIN_BRIEF_Three_promises_and_perils_of_Big_D...
2017, 60% of Big Data projects will fail to go beyond piloting and experimentation and will be abandoned." Why is history repeating itself? It's not for lack of inter-

10/26 - Evaluating Big Data Projects - Success and Failure U...
integralleadershipreview.com/10945-evaluating-big-data-projects-succes...
With the increasing focus on information analytics and "big data," the risks of lagging or failing projects are rising due to the complexity of the initiatives and the ...

Why the Majority of Big Data Projects Fail - Qubole
<https://www.qubole.com/blog/big-data/big-data-projects-fail/>
Jun 18, 2015 - To truly experience growth in the future, most businesses are turning to big data. In many cases, big data is seen as the new trend guaranteed ...

The Data Economy: Why do so many analytics projects fail?
analytics-magazine.org/the-data-economy-why-do-so-many-analytics-pr...
Key considerations for deep analytics on big data, learning and insights. ... Accord' Gartner, more than half of all analytics projects fail because they aren't ...

Where Big Data Projects Fail - Data Science Central
www.datasciencecentral.com/profiles/blogs/where-big-data-projects-fail
May 11, 2015 - In fact, I predict that half of all big data projects will fail to deliver against their expectations. Failure can happen for many reasons, however ...

Study reveals that most companies are failing at big data | CIO
www.cio.com/.../big-data/study-reveals-that-most-companies-are-failing-at-big-data.ht...
Nov 10, 2015 - Research from PwC and Iron Mountain reports some surprising statistics about how companies are using the data they collect.

Where Big Data Projects Fail - Forbes
www.forbes.com/sites/bernardmarr/2015/03/17/where-big-data-projects-fail/
Mar 17, 2015 - Over the past 6 months I have seen the number of big data projects go up significantly and most of the companies I work with are planning to ...

8 Reasons Big Data Projects Fail - InformationWeek
www.informationweek.com/big-data/big-data.../8...big-data-projects-fail/a/.../129784...
Aug 7, 2014 - Most companies remain on the big data sidelines too long, then fail. An iterative, start-small approach can help you avoid common pitfalls.

Four Reasons Why Big Data Analytics Projects Fail | Big Data Page by ...
bigdatapage.com/four-reasons-why-big-data-analytics-projects-fail-or-do-they/
2015 - Surveys say that many big data analytics projects end in failure. Four reasons why, question: Did they really fail?

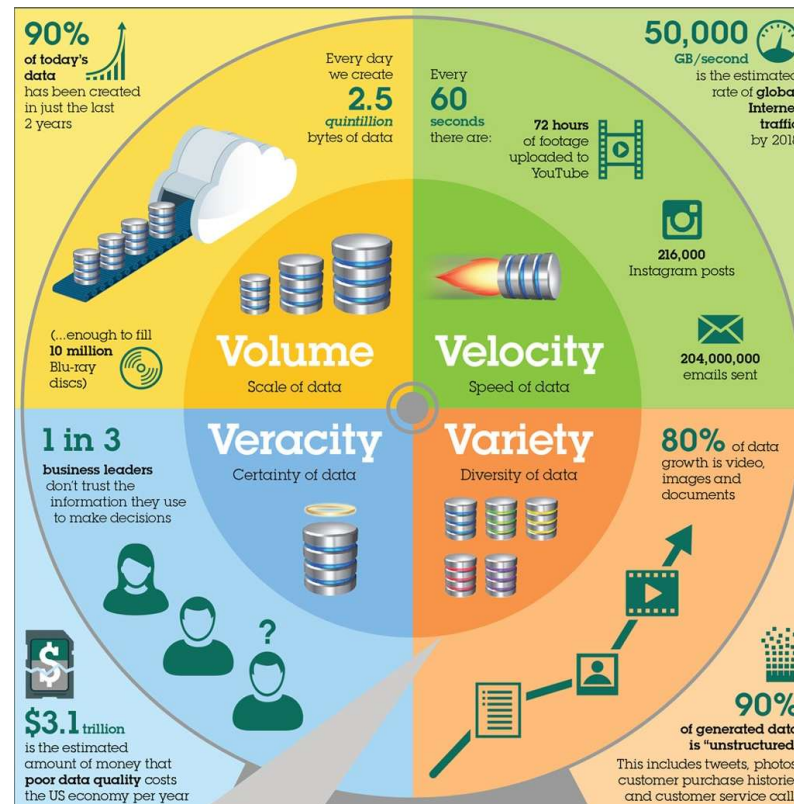
How to Prevent Big Data Analytics Failures - Smarter With Gartner
www.smarterwithgartner.com/how-to-prevent-big-data-analytics-failures/
2015 - Big data analytics projects don't fail for a single reason, nor due to technology alone.

Without a data-driven culture, big data projects will fail - SearchCIO
searchtarget.com/.../Without-a-data-driven-culture-big-data-projects-will-fail
Without a data-driven culture as vital to the success of any big data project.

Challenges

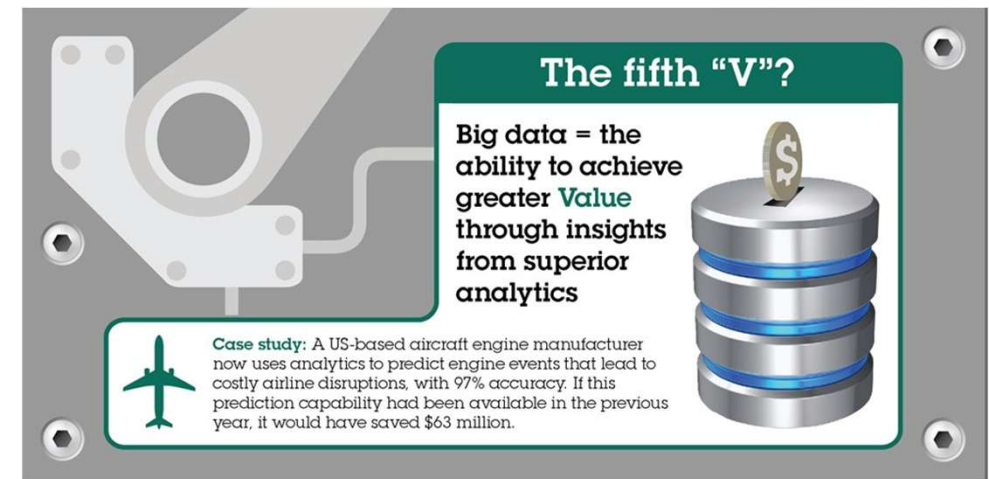
- Mapping the business question to a technique / setup (no one-size fits all)
- Quality/availability of data
- Realizing the amount of effort required in pre-processing
- Low amount of training data, either instances or features
- Predicting the future is hard (who'd have thought!)
- Incorporating domain knowledge, explaining models
- Strong validation / backtesting setup requires time and enough data

So what about big data?



<http://www.ibmbigdatahub.com/infographic/extracting-business-value-4-vs-big-data>

So what about big data?



<http://www.ibmbigdatahub.com/infographic/extracting-business-value-4-vs-big-data>

Our philosophy

- This programme is **all about techniques and applications**
 - Hands-on notebooks make the techniques concrete, but this is no tool or vendor training
- Start from YOUR business problem
- Do not buy *solutions* before you know and understand the problem

6. Tooling

Python: hands-on & Datacamp

- Python has become the universal language for machine learning and AI
- Through e.g. the concept of notebooks, it has become attractive even to non-technical users
- Code generation capabilities of GenAI-tools make for a much less steep learning curve
- Datacamp classroom
- Free online trainings in Python
- All content for free, for 6 months starting today
- Interesting tracks:
 - <https://www.datacamp.com/tracks/data-scientist-with-python> (90 hours, 23 courses)

Wifi access for today

- Network: iotd
- Password: kW?TgJD3
- Once you received your student ID card, you can login into campusroam or eduroam networks with your student ID and password

